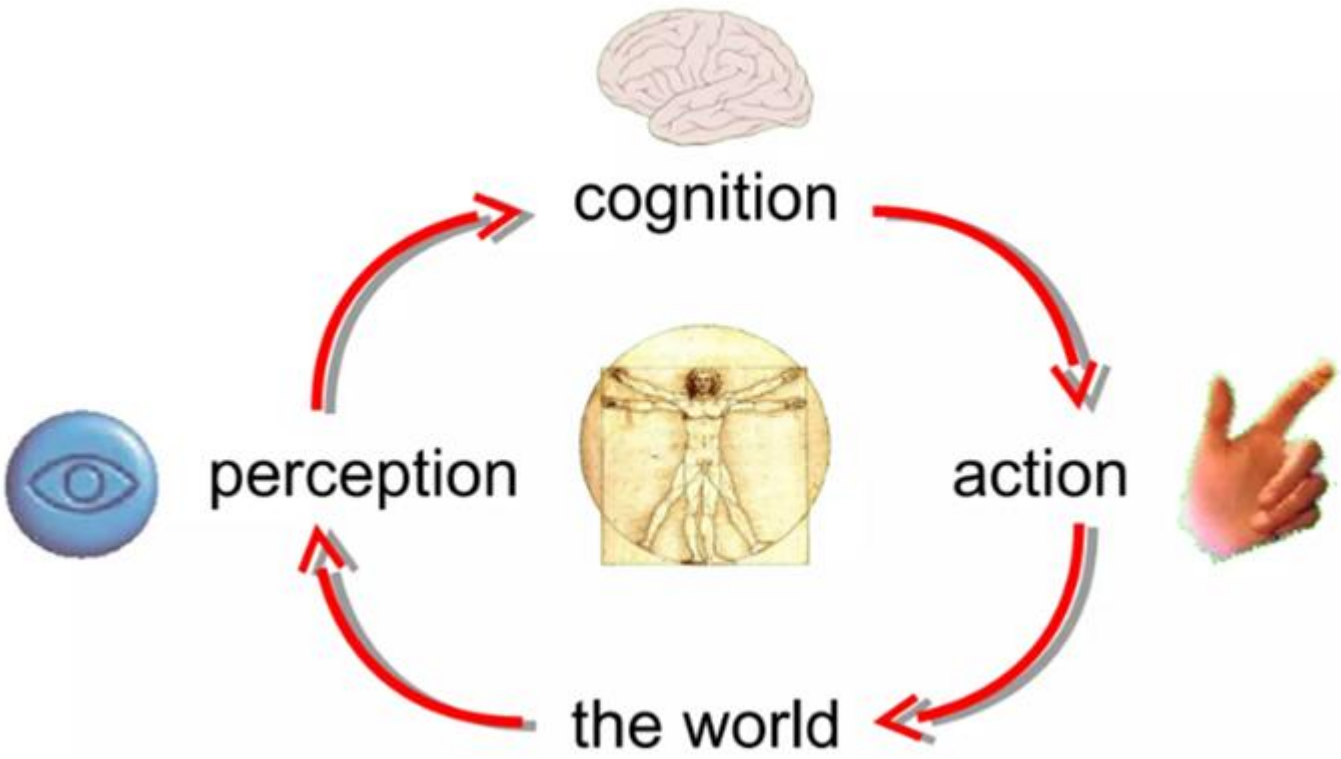
Artificial Intelligence

Big picture

- Human Agents
- AI Agents
- Fuel/Enablers for AI
- Enterprise AI
- Course Curriculum

Human Agent



- We live & act autonomously

Perception

- Sensory organs captures the input and perceived/ interpreted by mind.
    - Sight (Eyes)
    - Sound (Ears)
    - Smell (Nose)
    - Taste (Tongue)
    - Touch (Skin)

Cognition

- Thinking
  - Learning -> Knowledge
  - Reasoning -> Decision Making
  - Imagination

- Feeling
  - Feelings
  - Emotions

- Memory

Action

- We perform the actions to be executed by mind in the world using our action organs
  - Speech (mouth)
  - Grasping (hands)
  - Locomotion (legs)
  - Reproduction (genitals)
  - Excretion

Artificial Intelligence

- A discipline of computer science used to create intelligent agents

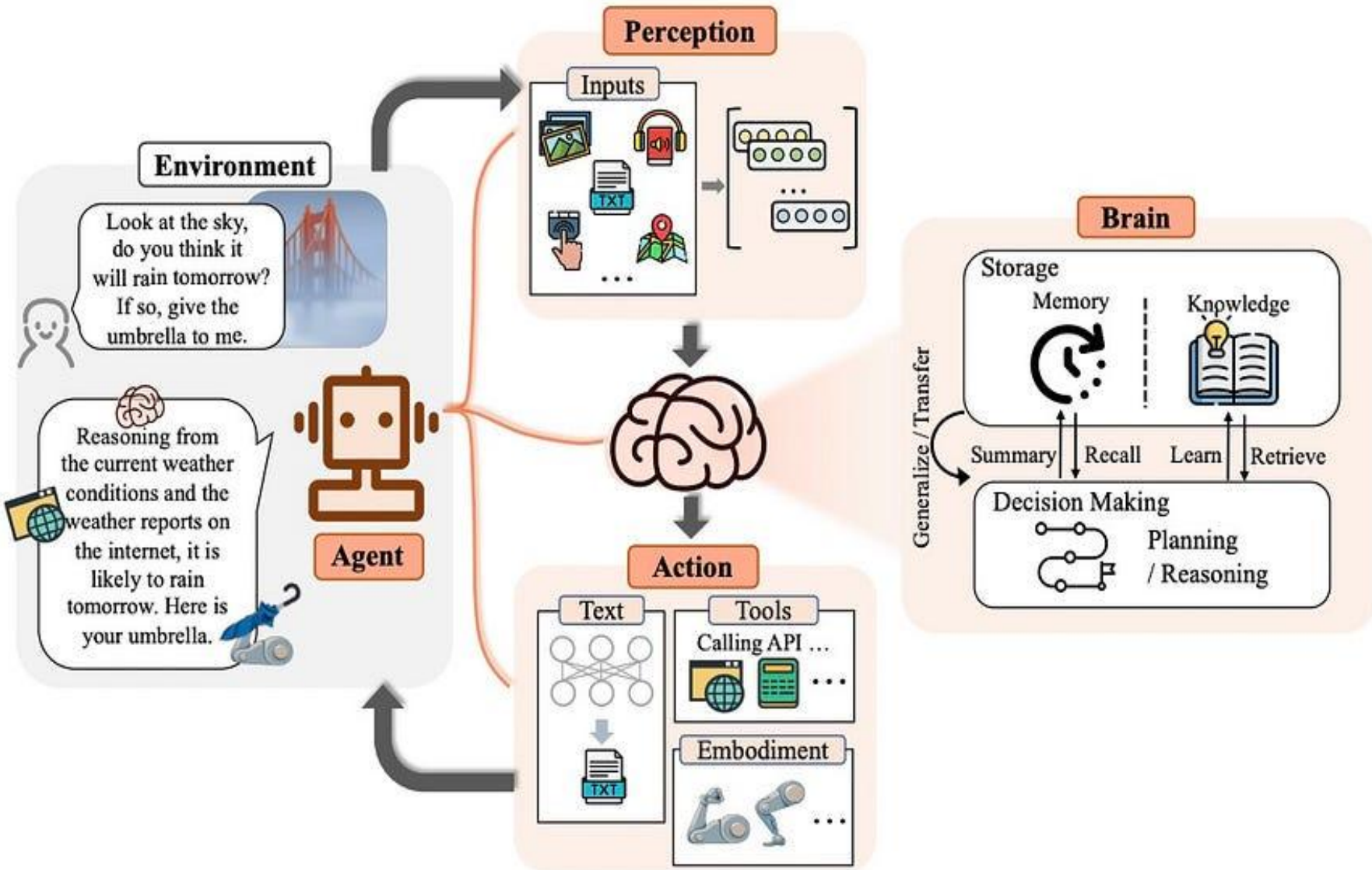
- Intelligent Agents are systems that can learn, reason and act autonomously

What is Generative AI?

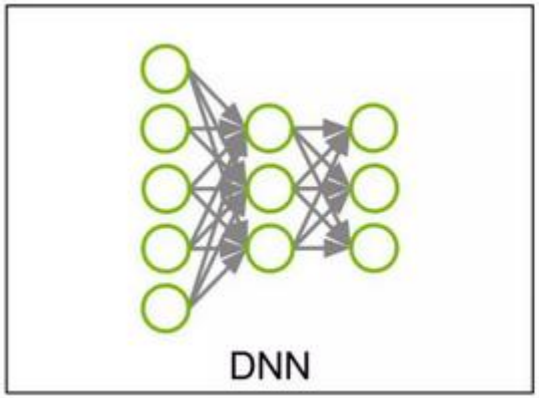
- A set of AI methodologies that can produce novel content that resembles the training data they were exposed to.

- Content type: text, code, images, music, video, etc.,

AI Agents

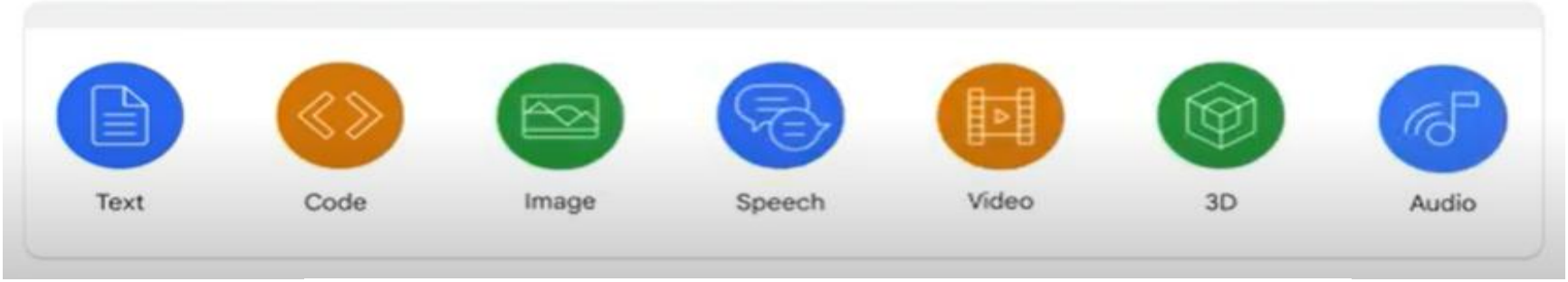

Perception
Inputs
TXT
Environment
Look at the sky, do you think it will rain tomorrow? If so, give the umbrella to me.
Reasoning from the current weather conditions and the weather reports on the internet, it is likely to rain tomorrow. Here is your umbrella.
Agent
Brain
Storage
Memory
Knowledge
Generalize / Transfer
Summary
Recall
Learn
Retrieve
Decision Making
Planning / Reasoning
Action
Text
TXT
Tools
Calling API ...
Embodiment

Enablers for AI Agents



- BIG DATA
- Brain Mathematical Models like Deep Neural Networks
- Computing Hardware Innovations like GPU, TPU, etc.,

BIGDATA (Multimodal)

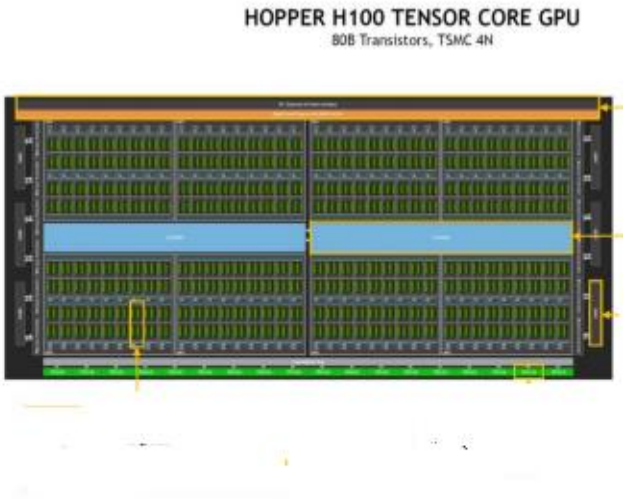


GPT-3 training data

Dataset	# tokens	Proportion within training
Common Crawl	410 billion	60%
WebText2	19 billion	22%
Books1	12 billion	8%
Books2	55 billion	8%
Wikipedia	3 billion	3%

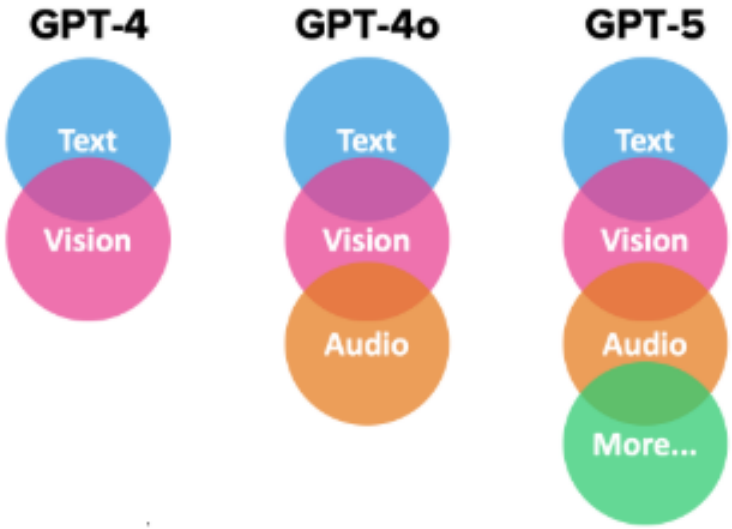
GPU

- NVIDIA H100 Architecture
  - 80B transistors
  - >15K cores
  - Optimized for parallel multiply/add operations
  - >1 TeraFLOPS at 16bit precision
  - >900 GB/s bandwidth



Large Language Models (LLM)

Large Multimodal Models (LMM)



Gemini

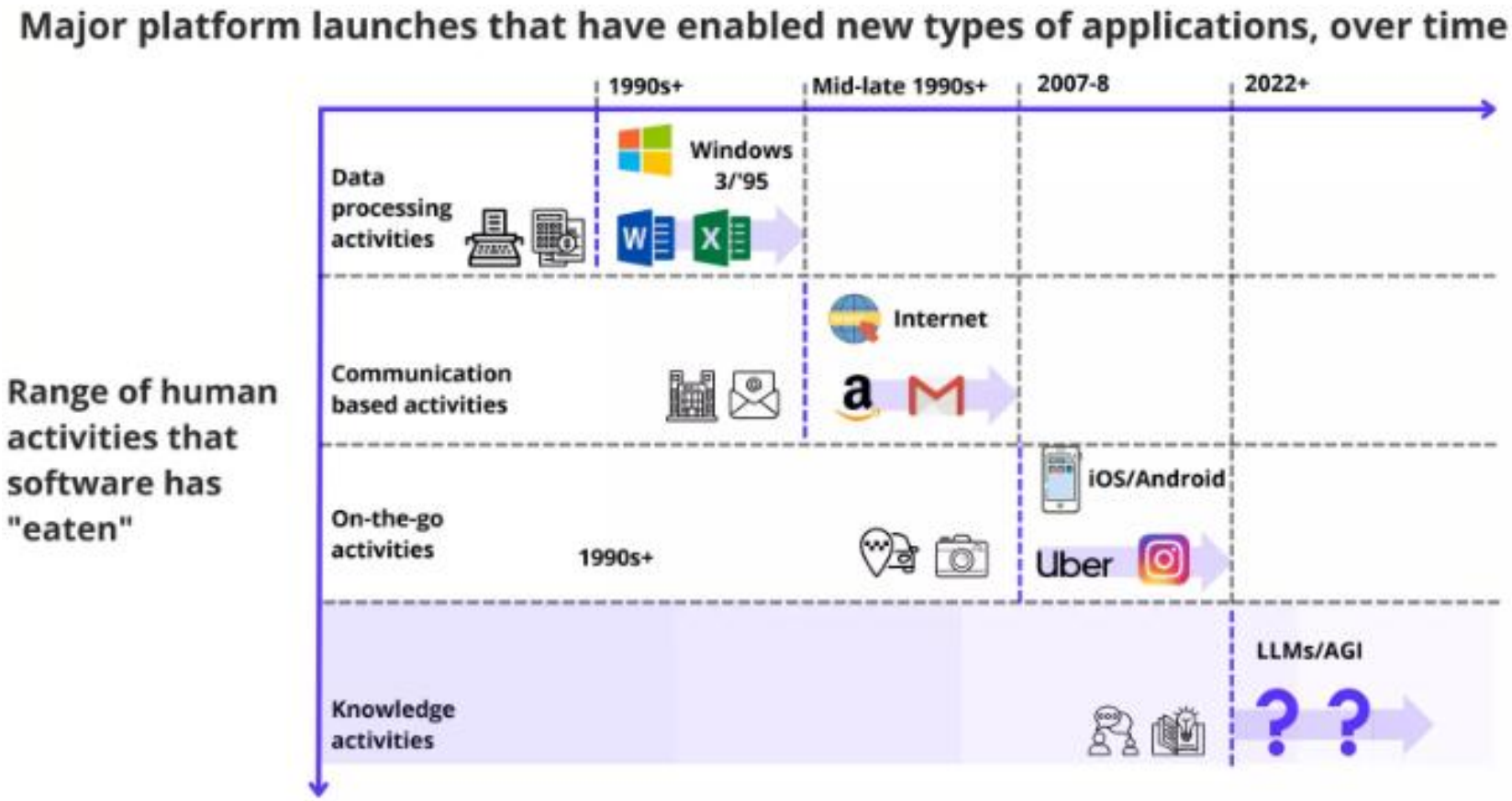
Large Reasoning Models (LRM)

DeepSeek-R1

OpenAI o3-mini

Gemini 2.0 Flash Thinking

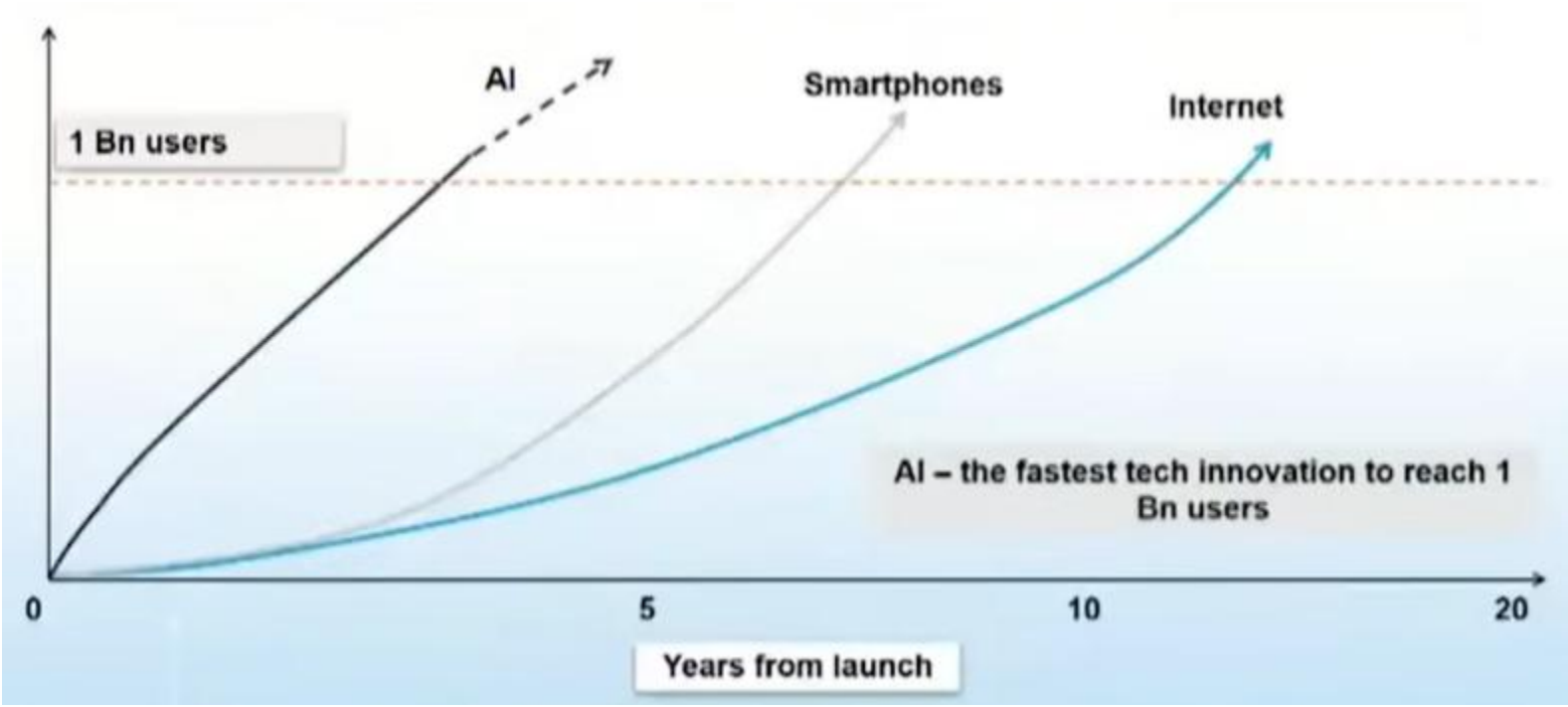
Enterprise AI: activity automation



LLMs are the Engine & AI agents are the Product

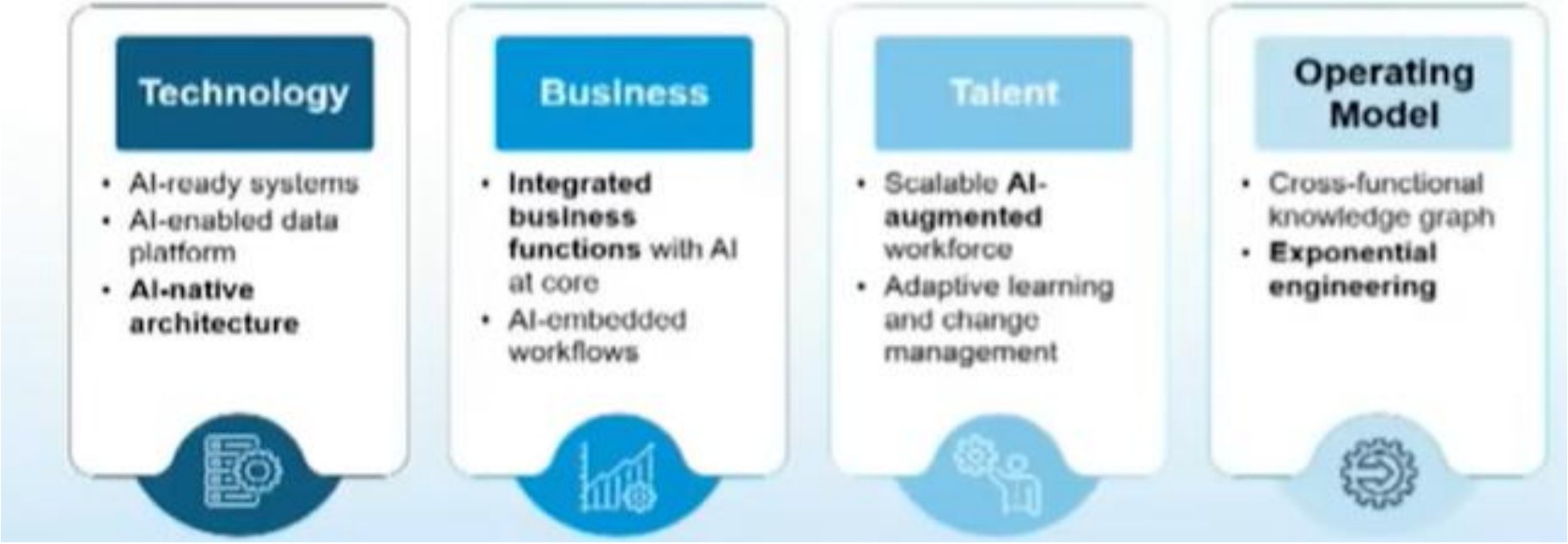
Enterprise AI: tech adoption time

- AI adoption is faster than earlier tech adoptions



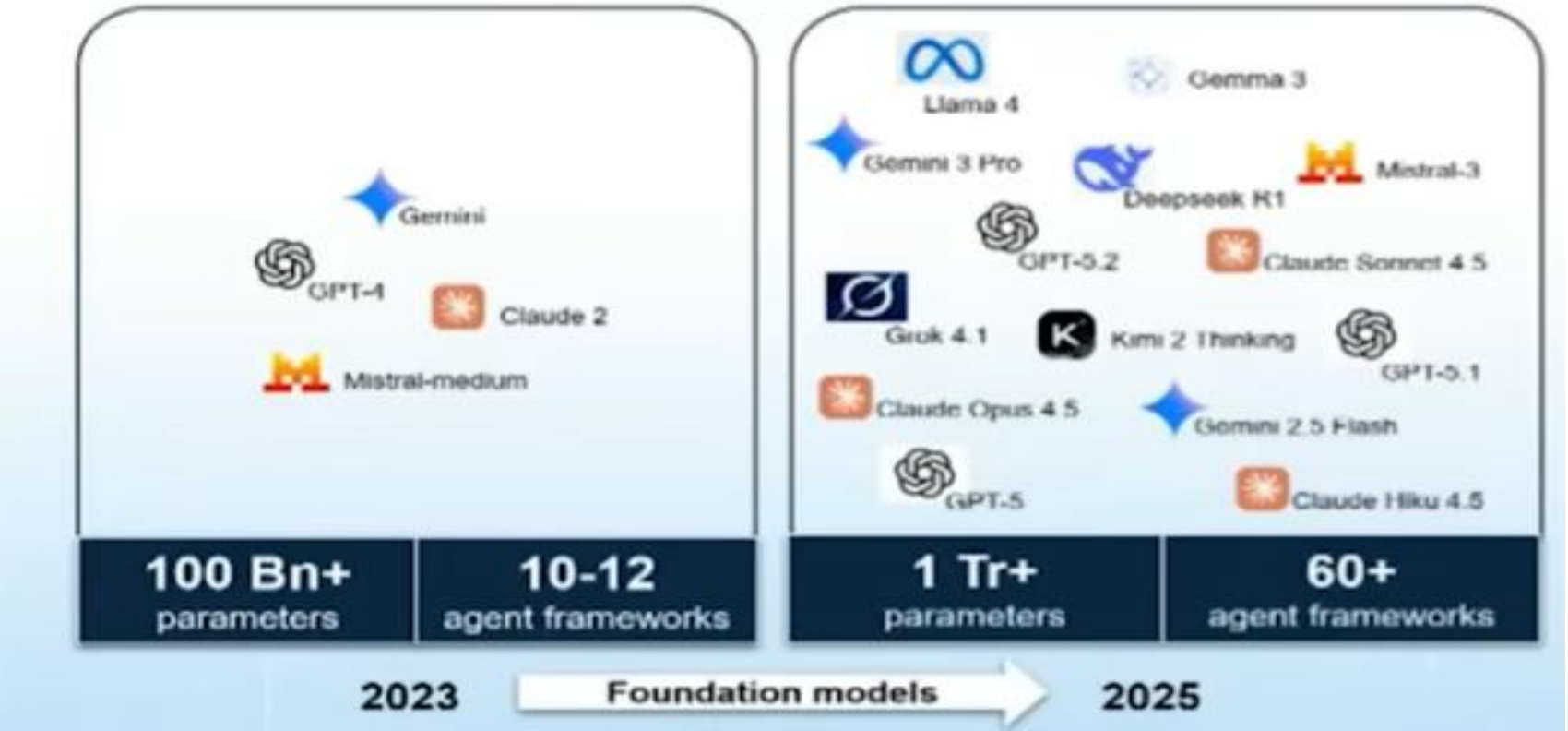
Enterprise AI: Multiple dimensions

- AI transformation is not a lift and shift. It requires a fundamental root surgery.



Enterprise AI: Tech Progress vs Value

- AI is evolving at astonishing speed led by rapid R&D, large capital access, hyper competitive market



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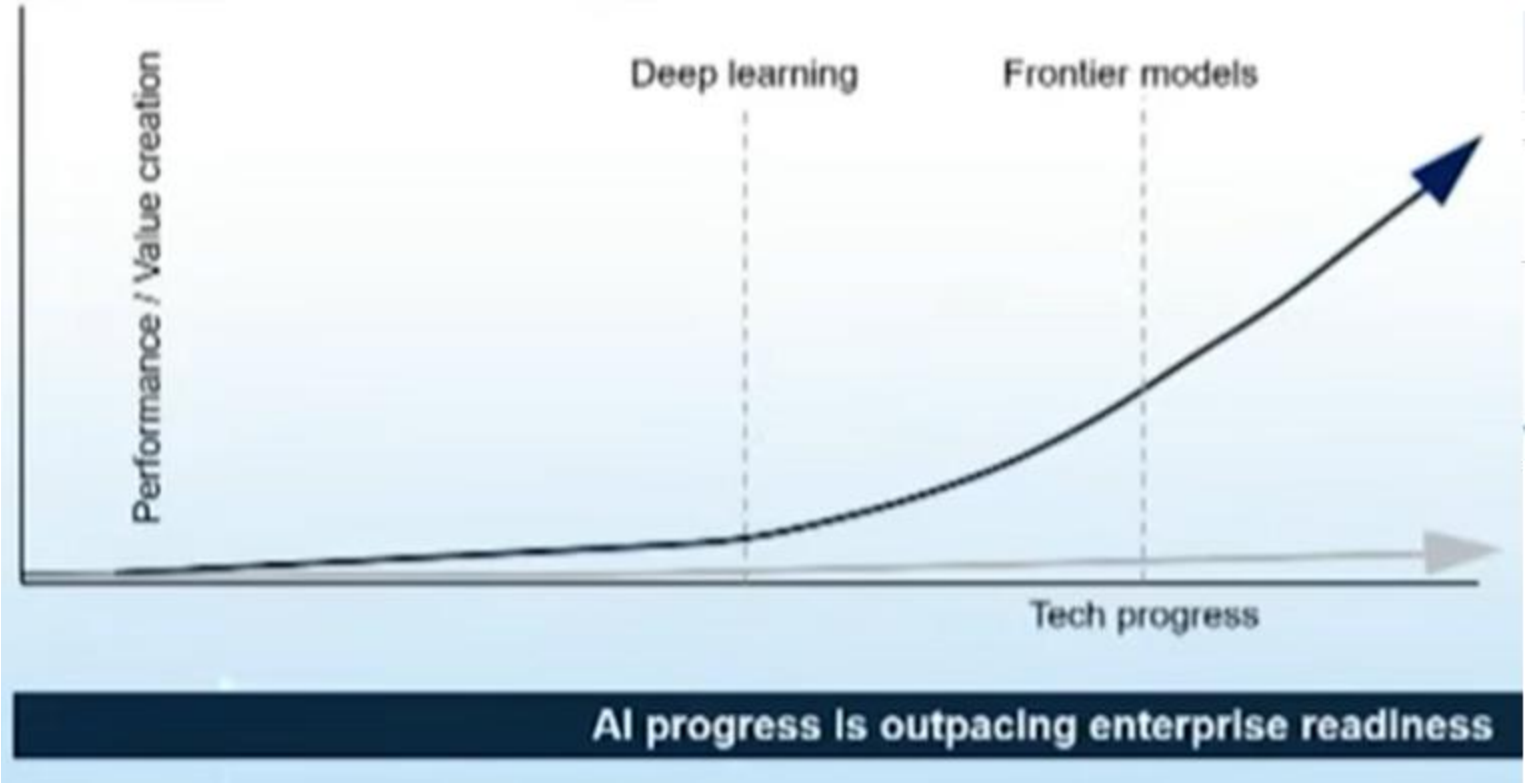
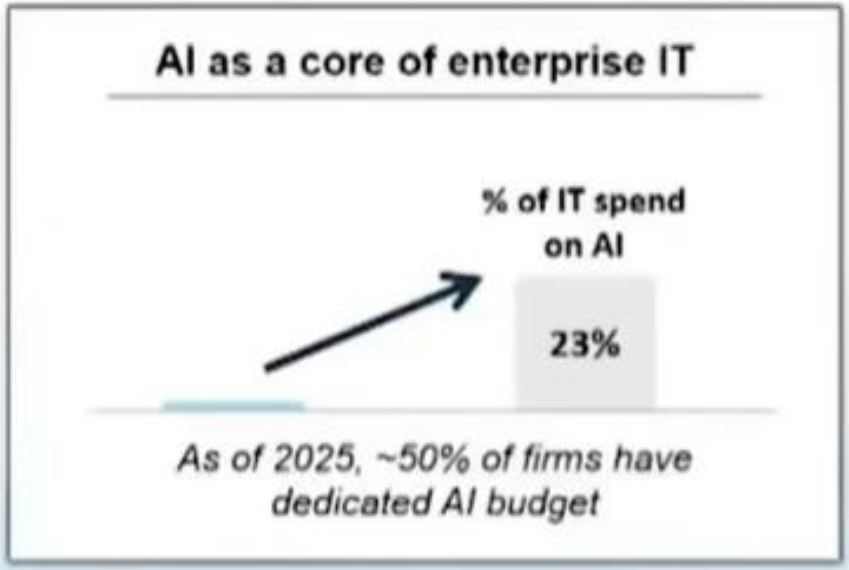


Enterprise AI: Job roles

- Talent demand is pivoting from legacy roles to high-growth AI skills

Enterprise AI: Build vs Buy

- Balance move towards build and re-engineering



Build	Buy
Customizable	Standardized
Proprietary	Vendor dependency
Organic and steady	Faster deployment
High internal control	External control
Continuous investment	Lock-in & renewals

Enterprises prefer proprietary agentic layer on top of the foundational models — building customizable to composable solutions

AI Impact: Across industry and business functions

Industry

- Healthcare
- Pharma & Drug discovery
- Education
- Public Administration
- Banking
- Media & Entertainment
- Gaming
- Fashion & Design
- Software Engineering

Business functions

- Customer Support
- HR & Recruitment
- Marketing & Sales
- Finance & Legal
- Logistics
- IT